

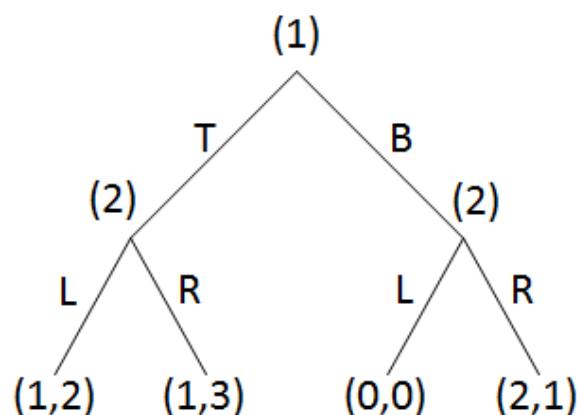
Section 21 Solution

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2 Exercises

2.1 Dynamic game

Consider the game below:



1. How many subgames does it have?

Answer: This game has three subgames (the two games with just player 2, and the entire game).

2. How many pure Nash equilibria? (pure strategy profiles that are mutual best responses)

Answer: We don't have a technique to identify the Nash equilibrium (as we have for the subgame perfect Nash equilibrium: *backward induction*), so we have to check all the potential strategy profiles. First, let's define the strategy sets of each player:

$$S_1 = \{T, B\}$$

$$S_2 = \{(L, L'), (L, R'), (R, L'), (R, R')\}$$

Now, let's check each potential strategy profile. It's enough to find one deviation that would make one

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player better off to discard the strategy profile as a Nash equilibrium.

- $(T, (L, L')) \notin NE$. Player 2 would be better off playing R instead of L.
- $(T, (L, R')) \notin NE$. Player 2 would be better off playing R instead of L.
- $(T, (R, L')) \in NE$. No player has a profitable deviation.
- $(T, (R, R')) \notin NE$. Player 1 would be better off playing B instead of T.
- $(B, (L, L')) \notin NE$. Player 1 would be better off playing T instead of B.
- $(B, (L, R')) \in NE$. No player has a profitable deviation.
- $(B, (R, L')) \notin NE$. Player 1 would be better off playing T instead of B.
- $(B, (R, R')) \in NE$. No player has a profitable deviation.

Nash equilibrium are:

$$NE = \{(T, (R, L')), (B, (L, R')), (B, (R, R'))\}$$

3. If this was a *simultaneous* game, how many Nash equilibrium would it have?

Answer: We are going to assume that playing L and L' are both “playing left”, and R and R' are both “playing right”. We can show that if this was a *simultaneous* game, the unique Nash equilibrium would be (B, R) :

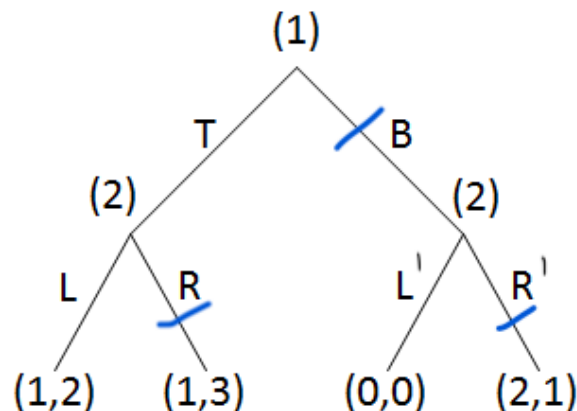
		(2)	
		L	R
(1)	T	<u>1</u> , <u>2</u>	1, <u>3</u>
	B	0, 0	<u>2</u> , <u>1</u>

4. Using backward induction, find the subgame perfect Nash equilibrium of the game.

Answer: Solving from the bottom to the top:

- Player 2 would choose R over L.
- Player 2 would choose R' over L' .
- Knowing what player 2 would do, player 1 would choose B over T.

The subgame perfect Nash equilibrium is $(B, (R, R'))$.



2.2 Cournot vs Stackelberg

Using the same setting from last section. Let's consider a market for cars with two firms with identical cost function given by $C(q) = 2q$. The market inverse demand function is given by $P(Q) = 9 - Q$.

1. What will the Cournot equilibrium output for each firm be?

Answer:

We are now in a Cournot competition set-up. A duopolistic firm maximizes profits:

$$\max_{q_i} (9 - (q_i + q_{-i})) q_i - 2q_i.$$

It follows that the first order conditions for the two firms are

$$9 - 2q_1^* - q_2^* - 2 = 0 \qquad 9 - 2q_2^* - q_1^* - 2 = 0$$

Solving this system of two equations we will get

$$q_1^* = q_2^* = \frac{7}{3}$$

Not surprisingly, the quantities produced by firms 1 and 2 are equal.

The total market output $Q^* = 2 * 7/3 = 14/3$.

2. What is the Stackelberg equilibrium output for each firm if firm 2 enters second?

Answer:

In the Stackelberg case, firm 1 maximizes taking into account the reaction function of firm 2. The reaction function of firm 2 is $BR_2(q_1) = q_2^*(q_1) = (7 - q_1)/2$. Therefore, firm 1 maximizes

$$\begin{aligned} \max_{q_1} (9 - (q_1 + BR_2(q_1))) q_1 - 2q_1 \\ \max_{q_1} (9 - (q_1 + (7 - q_1)/2)) q_1 - 2q_1 \end{aligned}$$

which leads to the first order condition

$$9 - 2q_1 - 7/2 + q_1 - 2 = 0$$

or

$$q_1^* = 7/2$$

and, using the reaction function of firm 2,

$$q_2^* = (7 - q_1^*)/2 = 7/4.$$

So the first mover produces twice as much. The total market output is $Q^* = 7/2 + 7/4 = 21/4$, larger than in Cournot.

3. How much profit will each firm make in the Cournot game? How much in Stackelberg?

Answer:

In the Cournot case, the profit for either firm is

$$\pi_C = (9 - 14/3) 7/3 - 14/3 = 91/9 - 14/3 = 49/9.$$

Profits in Stackelberg differ. The profit of the leader is

$$\pi_S^1 = (9 - 21/4) 7/2 - 14/2 = 105/8 - 14/2 = 49/8 > \pi_C,$$

while the profit of the follower is

$$\pi_S^2 = (9 - 21/4) 7/4 - 14/4 = 105/16 - 14/4 = 49/16 < \pi_C$$

4. Which type of market do consumers prefer: Cournot duopoly or Stackelberg duopoly? Why?

Answer:

Consumers prefer the market with the most total output (and by consequence the lowest price), since this maximizes consumer surplus. Formally, consumers maximize

$$\int_0^Q (D(p) - p) dq' = \int_0^Q ((9 - q') - (9 - Q)) dq' = [Qq' - q'^2/2]_0^Q = Q^2 - Q^2/2 = Q^2/2,$$

which is increasing in total quantity produced. (This integration was not required.) Therefore consumers prefer the Stackelberg duopoly, which has the highest total production.